

Def. A Homotopy of paths in  $X$  is a family

$$f_t: I \rightarrow X, \quad 0 \leq t \leq 1 \quad \text{such that}$$

- 1) For all  $0 \leq t \leq 1$ ,  $f_t(0) = x_0$  and  $f_t(1) = x_1$ .
- 2)  $F: I \times I \rightarrow X; (s, t) \mapsto f_t(s)$  is continuous.

Given two path  $P$  and  $g$  from  $x_0 \in X$  to  $x_1$ , they are said to be homotopic if there exists a homotopy of paths  $f_t$  satisfying  $P = f_0$  and  $g = f_1$ .

Denoted as  $P \simeq g$ .

Example. Linear Homotopy

Given two path  $P$  and  $g$  in  $\mathbb{R}^n$  having the same endpoint  $x_0$  and  $x_1$ , define

$$f_t: I \rightarrow \mathbb{R}^n; s \mapsto (1-t)P(s) + t g(s)$$

for each  $0 \leq t \leq 1$ . Then, given  $t$ ,

$$f_t(0) = (1-t)P(0) + t g(0) = x_0 \quad \text{and} \quad f_t(1) = (1-t)P(1) + t g(1) = x_1$$

(In actually, for any  $s \in [0, 1]$ ,  $P(s) = g(s) \Rightarrow f_t(s) = P(s)$ )

And, to show  $F: I \times I \rightarrow \mathbb{R}^n; (s, t) \mapsto f_t(s)$  is continuous, let  $M > 0$  be a bound for  $P$  and  $g$ .

Let  $(s_0, t_0) \in I \times I$  be given. Given  $\epsilon > 0$ , there is  $\delta > 0$  such that

$$\|s - s_0\| < \delta \Rightarrow \|f_{t_0}(s) - f_{t_0}(s_0)\| < \epsilon \quad \text{and}$$

$$\begin{aligned} \|t - t_0\| < \delta \Rightarrow \|f_t(s) - f_{t_0}(s)\| &= \|(1-t)P(s) + t g(s) - [(1-t_0)P(s) + t_0 g(s)]\| \\ &= \|(t_0 - t)P(s) + (t - t_0)g(s)\| \\ &\leq |t_0 - t| \cdot (\|P(s)\| + \|g(s)\|) < \epsilon. \end{aligned}$$

Now,

$$\|(s, t) - (s_0, t_0)\| < \delta \Rightarrow \|f_t(s) - f_{t_0}(s_0)\| \leq \|f_t(s) - f_{t_0}(s)\| + \|f_{t_0}(s) - f_{t_0}(s_0)\| < 2\epsilon.$$

So  $P \simeq g$   $\square$

**Prop.** Let  $X$  be a Topological Space. Given two points  $x_0$  and  $x_1$  in  $X$ , define an relation on the set of paths from  $x_0$  to  $x_1$  as:  $p \sim q \Leftrightarrow p \simeq q$ . Then, this relation is equivalence relation.

**Proof.** Reflexivity is evident, by  $f_t = f$ , the constant homotopy.

Symmetry is given by the inverse homotopy: if  $f_0 \simeq f_1$  via  $f_t$ , then  $f_1 \simeq f_0$  via  $f_{1-t}$ .

To show the Transitivity, suppose that  $f_0 \simeq f_1$  via  $f_t$  and  $g_0 \simeq g_1$  via  $g_t$ , with  $f_1 = g_0$ .

Define  $h: I \times I \rightarrow X: (s, t) \mapsto \begin{cases} f_{2t}(s) & \text{if } 0 \leq t \leq \frac{1}{2} \\ g_{2t-1}(s) & \text{if } \frac{1}{2} \leq t \leq 1. \end{cases}$

It is well-defined since  $f_1 = g_0$ . And, Continuity is given by the Pasting Lemma.

**Def.** A path  $p: I \rightarrow X$  is called a loop if  $p(0) = p(1) = x_0 \in X$ , and the point  $x_0$  is called the basepoint.

The collection of all homotopy classes  $[f]$  of loops  $f: I \rightarrow X$  at the basepoint  $x_0$ ,

$$\pi_1(X, x_0) \stackrel{\text{def}}{=} \{ [f] \mid f \text{ is a loop at } x_0 \}$$

is called the fundamental group under the operation

$$[f][g] \stackrel{\text{def}}{=} [f \cdot g] \text{ where } f \cdot g: I \rightarrow X: s \mapsto \begin{cases} f(2s) & \text{if } 0 \leq s \leq \frac{1}{2} \\ g(2s-1) & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases}$$

This  $\pi_1(X, x_0)$  is well-defined as a set clearly, and the operation is also well defined by: Suppose that  $[f_1] = [f_2]$  and  $[g_1] = [g_2]$ . Then  $f_1 \simeq f_2$  and  $g_1 \simeq g_2$  via  $\hat{f}_t$  and  $\hat{g}_t$ .

$$\begin{aligned} [f_1][g_1] &= [f_1 \cdot g_1] = [f_2 \cdot g_2] \text{ because } f_1 \cdot g_1 \simeq f_2 \cdot g_2 \text{ via } \hat{f}_t \cdot \hat{g}_t \\ &= [f_2][g_2] \end{aligned}$$

Moreover, the operation satisfies the Group Axiom, the proof lies in the next Page.

proof of the fundamental group.

First, given a path  $f: I \rightarrow X$ , define a reparametrization of a path  $f$  as a composition  $f \circ \varphi$  where  $\varphi: I \rightarrow I$  is any continuous map s.t.  $\varphi(0)=0$  and  $\varphi(1)=1$ .

The reparametrization preserves homotopy, that is,

$f \circ \varphi \simeq f$  via  $f \circ \varphi_t$  where  $\varphi_t(s) = (1-t)\varphi(s) + ts$ , so that  $\varphi_0 = \varphi$  and  $\varphi_1 = \text{id}$ .

**Associativity.** Let  $f, g, h$  be paths with  $f(1) = g(0)$  and  $g(1) = h(0)$ .

Then, the composition  $(f \cdot g) \cdot h = f \cdot (g \cdot h)$  are defined. Moreover,

$f \cdot (g \cdot h)$  is a reparametrization of  $(f \cdot g) \cdot h$ , because

$$f \cdot (g \cdot h)(s) = \begin{cases} f(2s) & \text{if } 0 \leq s \leq \frac{1}{2} \\ g(4s-2) & \text{if } \frac{1}{2} \leq s \leq \frac{3}{4} \\ h(4s-3) & \text{if } \frac{3}{4} \leq s \leq 1 \end{cases} \quad \text{and}$$

$$(f \cdot g) \cdot h(s) = \begin{cases} f(4s) & \text{if } 0 \leq s \leq \frac{1}{4} \\ g(4s-1) & \text{if } \frac{1}{4} \leq s \leq \frac{1}{2} \\ h(2s-1) & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases} \quad \text{therefore}$$

$$f \cdot (g \cdot h) = (f \cdot g) \cdot h \circ \varphi \quad \text{where} \quad \varphi(s) = \begin{cases} \frac{1}{2}s & \text{if } 0 \leq s \leq \frac{1}{2} \\ s - \frac{1}{4} & \text{if } \frac{1}{2} \leq s \leq \frac{3}{4} \\ 2s-1 & \text{if } \frac{3}{4} \leq s \leq 1 \end{cases}$$

so  $f \cdot (g \cdot h) \simeq (f \cdot g) \cdot h$ , the associative is proven

**Identity.** Given a path  $f: I \rightarrow X$ , let  $C$  be the constant path, as

$$C_i: I \rightarrow X: s \mapsto f(i) \quad (i=0,1)$$

Then  $f \cdot C_1$  is a reparametrization of  $f$  via

$$\varphi: I \rightarrow I: s \mapsto \begin{cases} 2s & \text{if } 0 \leq s \leq \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases}$$

and similarly  $C_0 \cdot f$  is also a reparametrization, so,  $C_0 \cdot f \simeq f \cdot C_1 \simeq f$ .

For a loop, the endpoints are fixed, so  $C(s) = x_0$  is two-sided identity in  $\pi_1(X, x_0)$ .

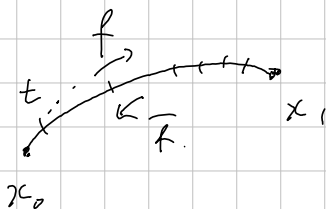
**Inverse.** Given a path  $f$  from  $x_0$  to  $x_1$ , define the inverse path

$$\bar{f}(s) = f(1-s) \quad \text{so that } \bar{f}(0) = f(1), \bar{f}(1) = f(0)$$

Then  $f \cdot \bar{f}$  is homotopic to a constant path  $C(s) = f(0)$ , via  $h_t = f_t \cdot \bar{f}_t$  where

$$f_t: I \rightarrow X: s \mapsto \begin{cases} f(s) & \text{if } 0 \leq s \leq 1-t \\ f(1-t) & \text{if } 1-t \leq s \leq 1 \end{cases}$$

similarly, for a loop,  $f \cdot \bar{f} \simeq \bar{f} \cdot f \simeq C(\equiv x_0)$ .



[Change of basePoint.

[Def. Let  $X$  be a space, and  $x_0, x_1 \in X$  be points contained a same path-component.

Let  $h: I \rightarrow X$  be a path from  $x_0$  to  $x_1$ , with the inverse curve  $\bar{h}(s) = h(1-s)$ .

Then, for any loop at  $x_1$ , the composition  $h \cdot f \cdot \bar{h}$  is a loop at  $x_0$ .

Rigorously,  $(h \cdot f) \cdot \bar{h} \neq h \cdot (f \cdot \bar{h})$  but  $(h \cdot f) \cdot \bar{h} \simeq h \cdot (f \cdot \bar{h})$ . Therefore, the map

$$\beta_h: \pi_1(X, x_1) \rightarrow \pi_1(X, x_0): [f] \mapsto [h \cdot f \cdot \bar{h}]$$

is well-defined, and is called a change of basepoint.

Moreover, for path-connected  $x_0$  and  $x_1$ , the map  $\beta_h$  is an Isomorphism.

Homomorphism:  $\beta_h[f \cdot g] = [h \cdot f \cdot g \cdot \bar{h}] = [h \cdot f \cdot \bar{h} \cdot h \cdot g \cdot \bar{h}] = \beta_h[f] \beta_h[g]$ .

Invertible:  $\beta_h$  satisfies

$$\beta_h \beta_{\bar{h}}[f] = \beta_h[\bar{h} \cdot f \cdot h] = [h \cdot \bar{h} \cdot f \cdot h \cdot \bar{h}] = [f].$$

and similarly  $\beta_{\bar{h}} \beta_h[f] = [f]$ .

[Def. A space  $X$  is called 'if it is path-connected and has trivial fundamental group.

[Def. Given a space  $X$ , a covering space of  $X$  consists of a space  $\tilde{X}$  and a map  $p: \tilde{X} \rightarrow X$  s.t.

For each  $x \in X$ , there is an open neighborhood  $U$  of  $x$  in  $X$  satisfying

$$p^{-1}[U] = \bigsqcup_{i \in I} \mathcal{O}_i, \text{ each } \mathcal{O}_i \text{ s.t. } \mathcal{O}_i \stackrel{\text{Homeo}}{\cong} U \text{ via } p.$$

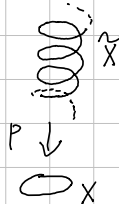
[Example. Given a space  $S^1$ , let  $p: \mathbb{R} \rightarrow S^1: s \mapsto (\cos 2\pi s, \sin 2\pi s)$ .

Let  $(1, 0) \in S^1$  be given. Then, for a neighborhood  $B_1(1, 0) \cap S^1$ ,

$$p^{-1}[B_1(1, 0) \cap S^1] = \bigsqcup_{n \in \mathbb{Z}} (n + (-\frac{1}{8}, \frac{1}{8}))$$

and for each  $n \in \mathbb{Z}$ ,

$$(n + (-\frac{1}{8}, \frac{1}{8})) \cong B_1(1, 0) \cap S^1 \text{ via } p.$$



Prop. Let  $X$  and  $Y$  are Path-Connected. Then,

$$\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y)$$

Proof. Notice that:  $f: Z \rightarrow X \times Y$  is continuous iff  $p_1 \circ f$  and  $p_2 \circ f$  both are continuous, where  $p_1: X \times Y \rightarrow X$  and  $p_2: X \times Y \rightarrow Y$  are Projections.

Therefore, there is a one-to-one Correspondence

$$\{f: I \rightarrow X \times Y \mid f \text{ is a loop at } (x_0, y_0)\}$$



$$\{g: I \rightarrow X \mid g \text{ is a loop at } x_0 \in X\} \times \{h: I \rightarrow Y \mid h \text{ is a loop at } y_0 \in Y\},$$

Under this Correspondence, we can define a bijective

$$\pi_1(X \times Y, (x_0, y_0)) \rightarrow \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

$$: [f] \mapsto ([g], [h]) \quad \text{where } f = (g, h)$$

is a bijective group Homomorphism, so proved

The Fundamental Group of the Circle.

Thm.  $\pi_1(S^1)$  is an infinite cyclic group generated by  $[w]$ , where

$$w(s) = (\cos 2\pi s, \sin 2\pi s) \text{ based at } (1, 0).$$

Note that  $[w]^n = [w_n]$  where  $w_n(s) = (\cos 2\pi ns, \sin 2\pi ns)$  for each  $n \in \mathbb{N}$ .

Consider  $p: \mathbb{R} \rightarrow S^1; s \mapsto (\cos 2\pi s, \sin 2\pi s)$  and  $\tilde{w}_n: \mathbb{I} \rightarrow \mathbb{R}; s \mapsto ns$ .

Then  $p$  is a covering map, and  $\tilde{w}_n$  is called the lift of  $w_n$ , satisfying

$$w_n = p \circ \tilde{w}_n.$$

Proof. Let  $f: \mathbb{I} \rightarrow S^1$  be a loop at the basepoint  $x_0 = (1, 0)$ .

Since  $S^1$  is path-connected,  $[f] \in \pi_1(S^1)$ .

Thm. Let  $X$  be a space and let  $w: I \rightarrow X$  be a path from  $x_0$  to  $x_1$ .  
Then, a map  $\hat{w}: \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$   
 $[\alpha] \mapsto [\bar{w} * \alpha * w]$

is a Group Isomorphism.

Thm. Let  $f: X \rightarrow Y$  be a continuous map such that  $f(x_0) = y_0$ .  
Then,  $f$  induces a group homomorphism

$$f_{\#}: \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0); [\alpha] \mapsto [f \circ \alpha].$$

Proof. Notice that given a loop  $\alpha: I \rightarrow X$  at  $x_0$ , a composition  $f \circ \alpha: I \rightarrow Y$  satisfies  $(f \circ \alpha)(0) = (f \circ \alpha)(1) = f(x_0) = y_0$ , so  $f \circ \alpha$  is a loop at  $y_0$ .

Let  $[\alpha], [\beta] \in \pi_1(X, x_0)$ . If  $[\alpha] = [\beta]$ , then there is a Homotopy

$F: I \times I \rightarrow X$  s.t.  $F(x, 0) = \alpha(x)$  and  $F(x, 1) = \beta(x)$ , so that

$f \circ F: I \times I \rightarrow Y$  satisfies  $(f \circ F)(x, 0) = (f \circ \alpha)(x)$  and  $(f \circ F)(x, 1) = (f \circ \beta)(x)$

therefore,  $\alpha \stackrel{f \circ F}{\sim} \beta$  implies  $f \circ \alpha \stackrel{f \circ F}{\sim} f \circ \beta$ , thus  $[f \circ \alpha] = [f \circ \beta]$ .

It proves the  $f_{\#}$  is well-defined. Now,

$$\begin{aligned} f_{\#}([\alpha] * [\beta]) &= f_{\#}([\alpha * \beta]) = [f \circ (\alpha * \beta)] = [(f \circ \alpha) * (f \circ \beta)] = [f \circ \alpha] * [f \circ \beta] \\ &= f_{\#}([\alpha]) * f_{\#}([\beta]) \quad \square \end{aligned}$$

Thm. Let  $f: X \rightarrow Y$  be a Homotopy equivalence such that  $f(x_0) = y_0$ .  
Then, the induced map  $f_{\#}$  is an isomorphism.